

o4_Project_Map_and_Site_Inventory

Project: Kristal Farms — Labrador Coast Partner Package

Document type: Partner-facing map and site inventory brief

Version: vo.1 draft

Status: Working draft pending final KML/CSV validation

Primary geography: Labrador coast

First target: Nain, Nunatsiavut, Labrador

Replication logic: Hopedale, Makkovik, Postville, Rigolet, and other coastal-accessible Labrador communities where the same hydro/compute/heat model may be screened.

1. Purpose

This document explains the project map package and the site-selection inventory for Kristal Farms' Labrador coast opportunity.

The map package is designed to help partners quickly understand:

1. why the project starts with **Nain**;
2. which Labrador coastal features matter for site selection;
3. which features are included in the current map inventory;
4. which features are deliberately excluded from the partner package;
5. what data still needs validation before engineering, investment, permitting, or community decisions.

The map is not intended to be a final engineering drawing, permitting map, land-rights map, or construction layout. It is a partner-screening tool that organizes the current thesis: **use local coastal hydro, place modular compute near village heat users, export compute by fibre, reuse heat locally, and expand by replicating the model across coastal communities rather than by pursuing inland megaprojects.**

2. Partner-facing map package

The intended map/data package is:

o4_Project_Map_and_Site_Inventory.md

maps/

kristal_farms_LABRADOR_COAST_PROJECT_ONLY_colored.kml

kristal_farms_LABRADOR_COAST_PROJECT_ONLY_feature_inventory.csv

kristal_farms_LABRADOR_COAST_PROJECT_ONLY_validation.csv

Current status

File	Status	Use
kristal_farms_LABRADOR_COAST_PROJECT_ONLY_colored.kml	Pending final attachment/validation	Visual map layer for Google Earth or GIS viewers.
kristal_farms_LABRADOR_COAST_PROJECT_ONLY_feature_inventory.csv	Pending final attachment/validation	Structured inventory of mapped features.
kristal_farms_LABRADOR_COAST_PROJECT_ONLY_validation.csv	Pending final attachment/validation	Evidence and validation status for each mapped feature.

Until the KML/CSV files are final, this document should be read as the **map architecture and inventory specification**, not as a final map release.

3. Project map logic

The map should show only the Labrador coast project logic. It should not become a general northern hydro map, a Nunavik planning map, or a map of every possible dam or data center concept.

The selected map logic is:

Labrador coast focus

- Nain as first target
- coastal-accessible hydro screening
- village-first compute pad siting
- heat users near the pad yard
- short MV electrical connection
- fibre export of compute output
- replication across other coastal communities

This means the map should prioritize features that help answer practical partner questions:

- Where is the first community target?
- Where is the candidate hydro resource?
- Is there plausible marine access?
- Can containers be placed near heat users?
- Can heat be delivered through short pipe runs?
- Is a short MV power connection plausible?
- Is fibre connectivity a required dependency?
- Which sites are included only as replication candidates?
- Which sites are excluded because they distract from the coastal community-scale model?

4. Included map layers

The KML should use clear, limited layers. Each layer should have a practical partner-facing purpose.

Layer 1 — Project geography

Purpose: Define the geographic scope of the partner package.

Include:

- Labrador coast project boundary or focus corridor;
- Nunatsiavut / Labrador coastal community context, where appropriate and validated;
- Nain as the first target community;
- replication communities for later screening.

Do not include broad Canada-wide or global strategy layers in this partner map.

Layer 2 — First target: Nain

Purpose: Make Nain visually and strategically central.

Include:

- Nain community marker;
- Nain harbour / marine access marker, if validated;
- possible village-edge or port-edge compute pad screening zone;
- nearby public heat users to be validated;
- possible greenhouse heat-use area to be validated.

Nain should be marked as:

Priority 1 — first target / site-confirmation candidate

The map should not imply final land control, final pad location, final permits, or final community approval.

Layer 3 — Candidate hydro resource

Purpose: Show the current hydro screening basis.

Include:

- Fraser River / Nain-area hydro candidate marker or screening corridor;

- hydro intake / powerhouse concepts only if sourced and clearly preliminary;
- hydro-to-village short MV connection concept line, if available;
- evidence status for capacity estimates and field-validation needs.

Current source basis identifies Fraser River near Nain as a potential community-scale hydro candidate in the approximate 15–20 MW range. This should be labeled as **preliminary source basis — not bankable engineering data**.

Recommended map label:

Fraser River / Nain Hydro Candidate

Status: preliminary screening

Capacity: source basis indicates ~15–20 MW; requires hydrology, geotechnical, environmental, Indigenous governance, and engineering validation

Role: potential local hydro supply for village-first compute/heat model

Layer 4 — Electrical connection concept

Purpose: Show the preferred infrastructure logic: avoid long high-voltage transmission.

Include:

- candidate short medium-voltage connection from hydro source to village substation;
- village substation or energy hub placeholder;
- compute pad feeder concept;
- metering handoff points, if available at concept level.

Recommended label:

Short MV Feed Concept

Purpose: connect local hydro to village energy hub / compute pad yard without long HV transmission build-out

Status: concept only; route and voltage class require utility and engineering review

Do not include speculative long HV corridors or inland transmission extensions.

Layer 5 — Compute pad yard

Purpose: Show where modular compute containers would logically sit.

The current design logic is village-first: data containers should be near heat users, not remote at the dam.

Include:

- candidate compute pad yard screening zone;
- port-edge or village-edge logistics access, if validated;
- fencing / controlled access area concept;
- distance to heat users and fibre access points, if known.

Recommended label:

Compute Pad Yard Screening Zone

Purpose: modular black-box compute containers near heat users and logistics access

Status: screening zone only; final site requires land, geotechnical, community, permitting, fibre, utility, and thermal-loop validation

Layer 6 — Heat users and thermal loop

Purpose: Show that heat reuse is not a side benefit; it is part of the site-selection logic.

Include candidate or validated heat users such as:

- public buildings;
- homes or housing clusters;
- school / clinic / community buildings, if validated;
- greenhouse site or greenhouse screening area;

- thermal storage location, if applicable;
- short district heat loop concept.

Recommended label:

Heat Reuse Candidate

Priority: public buildings and homes first; greenhouse as seasonal/summer sink

Status: candidate only unless confirmed by building audits and community process

Do not map private buildings as confirmed heat offtakers unless consent and validation exist.

Layer 7 — Cooling source / rejection safeguard

Purpose: Show the heat-first cooling hierarchy and environmental safeguards.

Include:

- seawater/bay non-contact heat exchanger concept location, if validated;
- dry-cooler backup zone, if applicable;
- no-contact environmental exchange note;
- exclusion note for small rivers as primary heat sink.

Recommended label:

Cold Source / Backup Heat Rejection Concept

Rule: reuse → store → reject

Design: non-contact plate heat exchange; IT loop and building/environment loops remain sealed and separate

Status: requires environmental, marine, thermal, corrosion, and permitting review

Layer 8 — Fibre / connectivity dependency

Purpose: Show that the project exports compute, not electricity.

Include:

- known or candidate fibre landing / network point, if validated;
- NOC / network handoff placeholder;
- fibre dependency note;
- redundancy requirement or path-protection note, if known.

Recommended label:

Fibre / Network Dependency

Purpose: export compute results and tenant traffic, not bulk electricity

Status: availability, latency, redundancy, service provider, and SLA must be validated

Do not imply tenant data visibility by the host. The map should show physical network dependency only.

Layer 9 — Replication communities

Purpose: Show expansion logic without diluting the Nain-first pitch.

Include markers for:

- Hopedale;
- Makkovik;
- Postville;
- Rigolet;
- other Labrador coastal locations only if relevant and sourced.

Recommended label:

Replication Screening Community

Status: future screening only

Role: potential Labrador coast replication candidate if hydro, heat, fibre, logistics, governance, and community value conditions are met

These communities should not be presented as selected sites unless separate due diligence has been completed.

5. Excluded map layers

The partner-facing map should deliberately exclude distracting or non-canonical layers.

Excluded layer	Reason for exclusion
Churchill Falls / inland mega-hydro project layers	The Labrador coast package is not pursuing inland megaprojects or long-transmission strategies.
Broad Nunavik mega-river development layers	Useful as background only; distracts from Nain-first Labrador coast focus.
Global AI infrastructure layers	Belongs in archive/background, not this site inventory.
Manicouagan-specific proposal layers	Not part of the Labrador coast partner package.
Speculative financial zones or revenue areas	Not validated and not appropriate for a map inventory.
Unconfirmed private property / land access claims	Land status must be verified before mapping as available.
Final engineering alignments	This document is pre-feasibility / partner-screening level.
Sensitive cultural, ecological, or community sites	Should not be mapped externally unless approved through the proper community and governance process.

6. Why certain sites are removed or de-prioritized

The map inventory should use a transparent removal logic. A site can be removed, archived, or de-prioritized for any of the reasons below.

6.1 Inland / transmission-heavy sites

Sites are removed from the partner-facing package if they require large inland transmission build-out, new long high-voltage corridors, or major remote substations. This conflicts with the project thesis: **use local hydro near coastal communities and export compute by fibre instead of exporting electricity by long transmission.**

6.2 Mega-dam or reservoir-heavy concepts

Large hydro concepts that require major reservoirs, large flooding areas, or significant river-system transformation are not part of the current partner thesis. They may remain in the internal source archive as comparison material, but they should not be presented as part of the Labrador coast community-scale model.

6.3 Poor heat-use geography

Sites are de-prioritized if compute containers cannot be located near real heat users. A technically attractive hydro site is not enough. The heat-first model requires short, practical thermal distribution to public buildings, homes, or greenhouse loads.

6.4 Weak marine or seasonal logistics access

Coastal-accessible sites are preferred because the project relies on modular container logistics. Sites requiring major roadwork, repeated long-haul trucking, or complex inland access are weaker candidates.

6.5 Unclear fibre dependency

Compute can be powered locally, but the business model depends on data connectivity. Sites without a credible fibre path, network partner, redundancy plan, or acceptable latency profile should be marked as unresolved or de-prioritized.

6.6 Environmental or governance sensitivity

Sites are removed or held back if they overlap with protected areas, sensitive habitat, unresolved rights issues, or community concerns that have not been addressed. No site

should be advanced without a proper FPIC/community process and environmental review.

6.7 Inconsistent with Nain-first sequencing

A site may be technically interesting but still excluded from the first package if it distracts from the initial partner story. The first partner package should prove one clear model: Nain first, then Labrador coastal replication.

7. Site inventory structure

The feature inventory CSV should use a simple, auditable structure.

Recommended fields:

feature_id

feature_name

feature_type

community

region

latitude

longitude

geometry_type

priority_level

status

source_basis

validation_status

partner_relevance

included_in_kml

included_in_partner_package

exclusion_reason

next_evidence_needed

notes

Feature types

Use controlled feature types such as:

community

hydro_candidate

port_or_marine_access

compute_pad_screening_zone

heat_user_candidate

greenhouse_candidate

thermal_storage_candidate

village_substation_candidate

mv_connection_concept

fibre_or_network_point

cooling_source_candidate

dry_cooler_backup_zone

replication_screening_community

excluded_site

archive_reference

Priority levels

Use three priority levels:

Priority	Meaning
Priority 1	Nain-first features required to evaluate the initial pilot.
Priority 2	Labrador coast replication screening features.
Priority 3	Background, archive, or comparison features not included in the first partner package.

Validation status

Use standard validation statuses:

Status	Meaning
Confirmed	Supported by verified source, reviewed, and safe to use externally.
Preliminary	Source basis exists but requires validation before partner reliance.
Concept	Planning logic only; no validated site evidence yet.
To validate	Needs external evidence, partner input, or field work.

Status	Meaning
Excluded	Deliberately removed from partner package.
Internal only	May be useful, but not for external distribution.

8. Current working site inventory

The following inventory is a working partner-facing structure. Coordinates and geometry should be populated only from validated KML/CSV/GIS sources.

Feature ID	Feature	Type	Priority	Status	Partner relevance	Next evidence needed
KF-LAB-001	Nain	Community / first target	1	Preliminary / to validate	First site-confirmation candidate	Community process, heat-load audit, land/pad screening, fibre status, utility input.
KF-LAB-002	Fraser River / Nain hydro candidate	Hydro candidate	1	Preliminary	Possible local hydro supply for pilot model	Hydrology, flow seasonality, head, geotechnical, environmental,

Feature ID	Feature	Type	Priority	Status	Partner relevance	Next evidence needed
						cost, permits, Indigenous governance.
KF-LAB-003	Nain harbour / marine access	Port/logistics	1	To validate	Container delivery and modular logistics	Port capacity, sealift window, crane/lift limits, staging area, ice/seasonality.
KF-LAB-004	Nain compute pad screening zone	Compute pad	1	Concept	Village-first container yard near heat users	Land access, geotech, noise, security, zoning, community acceptance, fibre/power access.
KF-LAB-005	Nain public-building heat users	Heat users	1	To validate	Core community benefit and diesel	Building list, heating systems,

Feature ID	Feature	Type	Priority	Status	Partner relevance	Next evidence needed
					displacement	load profile, radiator temperatures, metering, owner consent.
KF-LAB-006	Nain greenhouse candidate	Seasonal heat sink	1	Concept	Summer/warm-month heat use and food value	Greenhouse operator, site, thermal load, economics, water, staffing, governance.
KF-LAB-007	Nain village substation / energy hub	Electrical node	1	Concept	Power handoff from hydro to pads and heat systems	Utility design, voltage, protection, metering, ownership, interconnection review.
KF-LAB-008	Short MV hydro-to-	Electrical route	1	Concept	Avoids long HV	Route survey,

Feature ID	Feature	Type	Priority	Status	Partner relevance	Next evidence needed
	village feed				transmission	land rights, cost, utility standards, environmental constraints.
KF-LAB-009	Fibre / NOC handoff	Connectivity	1	To validate	Required for compute export and tenant SLA	Provider, capacity, latency, redundancy, outage history, service agreement.
KF-LAB-010	Bay/seawater non-contact cold source	Cooling/rejection safeguard	1	Concept	Backup heat rejection and cooling resilience	Marine study, intake/rejection feasibility, corrosion design, thermal limits, permitting.
KF-LAB-011	Hopedale	Replication	2	Future screening	Labrador coast replication	Hydro/fibre/heat/logistics/co

Feature ID	Feature	Type	Priority	Status	Partner relevance	Next evidence needed
		community			n candidate	mmunity screening.
KF-LAB-012	Makkovik	Replication community	2	Future screening	Labrador coast replication candidate	Hydro/fibre/heat/logistics/community screening.
KF-LAB-013	Postville	Replication community	2	Future screening	Labrador coast replication candidate	Hydro/fibre/heat/logistics/community screening.
KF-LAB-014	Rigolet	Replication community	2	Future screening	Labrador coast replication candidate	Hydro/fibre/heat/logistics/community screening.
KF-LAB-015	Churchill Falls / inland mega-hydro	Excluded reference	3	Excluded	Not part of the current package	Keep as internal comparison only.
KF-LAB-016	Broad Nunavik mega-river sites	Archive/comparison	3	Internal only	Shows why community-scale/coastal model is preferred	Use only in hydro appendix or exclusion logic.

Feature ID	Feature	Type	Priority	Status	Partner relevance	Next evidence needed
KF-LAB-017	Manicouagan-specific concept	Archive/comparison	3	Internal only	Older proposal geography; not Labrador package	Keep in internal source archive.

9. Map legend

Recommended visual legend:

Symbol / color	Feature type	Meaning
Large star	First target	Nain first target.
Blue circle	Hydro candidate	Candidate hydro source or hydro screening area.
Blue line	MV connection concept	Conceptual short medium-voltage connection.
Orange square	Compute pad	Candidate modular container pad yard.
Red / heat icon	Heat user	Public building, housing cluster, greenhouse, or thermal load.
Green house icon	Greenhouse	Seasonal heat sink / food-production opportunity.

Symbol / color	Feature type	Meaning
Purple line / point	Fibre / NOC	Connectivity dependency or handoff.
Anchor icon	Marine logistics	Port, harbour, or sealift staging feature.
Grey marker	Excluded / archive	Not part of partner-facing first package.
Dashed outline	Screening zone	Conceptual area, not final site control.

The KML should avoid visual clutter. Nain-first features should be visually dominant. Replication communities should be visible but secondary.

10. Validation checklist

Before release to a partner, each mapped feature should be checked against this list.

Validation question	Required before external use?
Is the feature name correct?	Yes
Are coordinates accurate enough for partner screening?	Yes
Is the feature clearly marked as confirmed, preliminary, concept, or excluded?	Yes
Is the source basis identified?	Yes
Is any sensitive community, cultural, or ecological information removed or generalized?	Yes

Validation question	Required before external use?
Does the feature imply land control or approval that has not been granted?	Must not
Does the feature imply final engineering design?	Must not
Does the feature support the Nain-first Labrador coast thesis?	Yes
Is the feature needed for partner decision-making?	Yes
Is the next evidence needed clearly stated?	Yes

11. Recommended evidence package for site confirmation

The map inventory should lead to a focused evidence request, not a broad research exercise.

For Nain / first target

Required next evidence:

1. community engagement and FPIC pathway;
2. hydro resource confirmation for the Fraser River / Nain candidate;
3. preliminary environmental screening;
4. port and sealift logistics review;
5. candidate pad-yard land screening;
6. short MV connection concept review with utility/engineering input;
7. fibre availability, redundancy, and SLA review;
8. heat-load audit for public buildings and housing clusters;
9. greenhouse feasibility and operator model;
10. permitting and governance map.

For replication communities

Required next evidence:

1. community interest and governance pathway;
 2. local or nearby hydro potential;
 3. practical village-first siting option;
 4. heat users within short thermal-loop distance;
 5. fibre availability;
 6. marine logistics access;
 7. environmental constraints;
 8. local economic and training opportunities.
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12. Partner use of this document

A partner should use this document to decide whether to proceed to a more detailed site-confirmation phase.

The immediate decision questions are:

1. Is the Nain-first site logic credible enough to begin formal site confirmation?
 2. Which partner can validate hydro, power connection, fibre, marine logistics, land, or community process?
 3. Which mapped features are strong enough to keep in the partner package?
 4. Which features should be moved to the internal archive?
 5. What evidence is required before investment-grade diligence?
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13. Source basis

This map inventory is based on the current Kristal Farms source corpus, especially:

- **Potentiel hydroélectrique isolé au Nunavik et au Labrador (≥ 15 MW)** – hydro screening basis and Labrador/Nain preliminary hydro candidate logic.
- **Kristal Farms – Heat Recycling Plan** – village-first siting, reuse → store → reject, two sealed circuits, non-contact exchange, public-building and greenhouse heat logic.
- **Kristal Farms – Cost Advantage & Strategic Rationale** – no long HV transmission, natural cold, heat as value, marine logistics, modular pads, export compute by fibre.

- **Kristal Farms Internal Reference Document** — technical architecture, local hydro integration, village substation, compute pads, heat loop, fibre/NOC, phasing.
 - **Documentation Kristal Farms** — black-box tenancy boundary, SLA, governance, FPIC, metrics and validation framing.
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14. Open items before v1 release

The following items must be completed before this file becomes a final partner-facing release:

- attach or generate the final Labrador-only KML;
 - attach or generate the final feature inventory CSV;
 - attach or generate the final validation CSV;
 - confirm whether Nain harbour, pad-yard, heat-user, fibre, and cold-source points can be shown externally;
 - remove or generalize sensitive community/environmental features;
 - review all labels for overclaiming;
 - confirm partner-facing status of Fraser River / Nain hydro estimate;
 - confirm source names and version dates;
 - align this file with `02_Labrador_Coast_Project_Thesis.md` and `05_Labrador_Coast_Hydro_Screening_Appendix.md` once drafted.
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15. Distribution status

Recommended distribution: partner-facing after review.

Do not distribute as final until: KML/CSV files are attached and validation statuses are reviewed.

Internal-only material: raw hydro tables, old Nunavik mega-project material, Manicouagan proposal layers, unresolved land/rights data, unvalidated private heat-user locations, and any sensitive community/ecological sites.